Monolithic Architecture

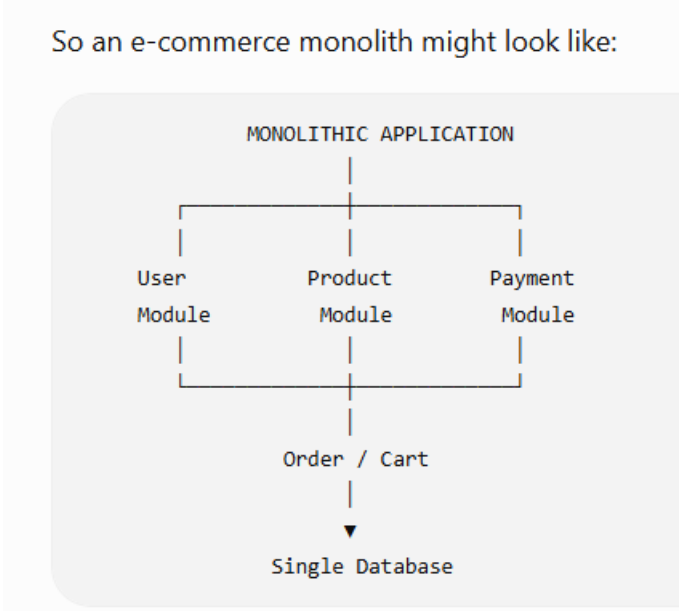
1. What is Monolithic Architecture

- **Monolithic Architecture** is a software architecture where the different parts/functions of an application are built and deployed as **one unified unit.**
- Think of it as **one big application containing everything.**
- **Monolithic = unified/single piece**
- A typical application may contain:
  - User Interface
  - Business Logic
  - Data Processing
  - Database Operations
  - User Functionality
  - Product Functionality
  - Payment Functionality
  - Order Functionality
  - Cart Functionality

2. What is a Module/Component?

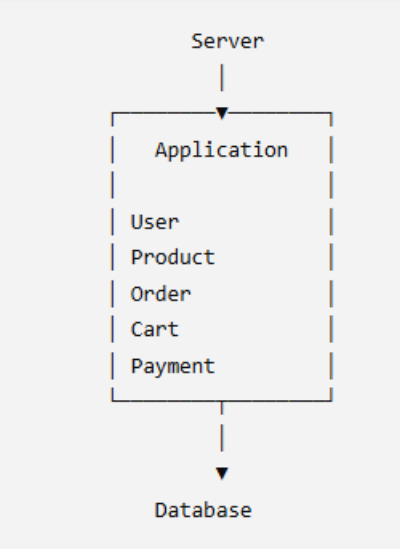
- A **module/component** is responsible for a particular functionality

So an e-commerce monolith might look like:



3. Single Server

- The monolithic application is also deployed as **one application unit on a server**



4. Modules are Interconnected and Interdependent

- Different components **communicate** with each other
- One component may depend on another component to perform its functionality.
- This leads to **tight coupling** between parts of the application
- So changes in one part of a large monolithic application can potentially affect other parts

For example:

```
Order Module
      |
      ├── depends on User Module
      |
      ├── depends on Product Module
      |
      └── depends on Payment Module
```

5. How Users Access a Monolithic Application

- The entire application is a single unit
- Client can access it through different interfaces

For example:



Clients could include:

- 🌐 Browser
- 🧪 Postman
- 📱 Mobile application
- Front-end application

6. The Big Picture

- **Monolithic** = **unified/single application**
- All major functionalities exist in **one codebase**
- Application contains multiple **modules/components**
- Modules may include User, Product, Order, Cart, Payment, etc.
- Modules are **interconnected and interdependent**
- This results in **tight coupling**
- Application is deployed as **one unit**
- Lecture's model uses a **single database**
- Application can be accessed through **browser, Postman, mobile app, etc.**
- This is the architectural style that the course is contrasting with **microservices**.

One-line mental model

> **Monolith = Many functionalities inside ONE big application.**